

Climbing Injury Prevention & Management

A Course for Physiotherapists

MODULE 5: TREATING & REHABILITATING CLIMBING INJURIES

5.1 Purpose of This Module

Module 4 established how to assess and differentiate the injuries most commonly seen in climbing patients. This module turns to treatment: evidence-based rehabilitation progressions for each site covered in Module 4, with particular emphasis on the tendon- and pulley-loading protocols that dominate climbing rehabilitation, climbing-specific exercise selection, and how to adapt standard physiotherapy loading principles to the specific movement patterns and demands of the sport. Module 6 then covers the final stage: grading readiness and managing the return to climbing itself.

5.2 General Principles of Tendon & Connective Tissue Rehabilitation

A large proportion of climbing injuries — pulley strain, flexor tenosynovitis, medial and lateral epicondylalgia, rotator cuff tendinopathy — are tendon or connective tissue conditions best understood through the tendon pathology continuum model: a spectrum from reactive tendinopathy through tendon dysrepair to degenerative tendinopathy, with treatment selected according to where the tissue sits on that spectrum rather than a single fixed protocol. This model, alongside evidence that high-load isometric contractions can reduce tendon pain without loss of muscle strength, underpins a staged loading approach that recurs throughout this module.

Stage	Loading Focus	Climbing-Specific Application
1. Isometric	Pain-modulating, low-technical-demand loading; sub-maximal holds within a tolerable pain level.	Gentle finger flexor holds or light hangs ("no-hangs") in an open-hand position, well within pain tolerance.
2. Isotonic / Concentric-Eccentric	Progressive resistance through range, building tissue capacity.	Light grip-strengthening work, controlled tendon glides, progressive forearm loading.
3. Energy Storage / Plyometric	Higher-rate loading that mimics the demands of real activity.	Controlled dynamic movement on the wall, progressing hold size and grip demand gradually.
4. Return to Sport	Full-intensity, sport-specific loading with monitoring.	Reintroducing full crimp, hangboard/campus training, and competition-level intensity last.

A pain-monitoring approach — permitting exercise and climbing-specific loading provided pain stays within an acceptable, non-worsening range — is a useful adjunct across most of the presentations in this module, allowing rehabilitation to proceed without requiring complete symptom resolution before reloading begins.

A practical way to apply this approach is to first identify the patient's **tolerable loading level** — the amount of force, duration, or climbing intensity that can be performed while keeping pain at **at or under 3/10** during the activity. Mild discomfort is often acceptable and does not necessarily indicate tissue damage, provided symptoms do not progressively worsen during the session. The second criterion is the **24-hour response**: any increase in pain or stiffness should settle back to the individual's baseline within **12–24 hours**. If either threshold is exceeded — pain greater than 3/10 during loading, or symptoms that remain elevated beyond 24 hours — the exercise or climbing load has likely been too high and should be reduced, not entirely stopped, just modified to fulfil the tolerable-load criteria.

Once a tolerable loading level has been established, rehabilitation should progress gradually by increasing one variable at a time — force, duration, repetitions, hold size, or climbing difficulty — while consistently remaining within these two pain-monitoring boundaries. This strategy promotes progressive tissue adaptation, maintains patient confidence, and reduces the likelihood of repeated overload or prolonged symptom flare-ups.

5.2.1 The Role of Manual Therapy in Climbing Injury Rehabilitation

Manual therapy and soft tissue techniques can be useful adjuncts to an active rehabilitation programme in climbers. They may be used to address pain, movement restriction, local tissue sensitivity, joint stiffness, altered neural mobility, or restrictions in surrounding tissues that may be contributing to impaired movement or loading.

As always, manual technique choice, dosage and grading should rely on thorough clinical reasoning and the findings of the individual assessment. Manual therapy should not be viewed as a substitute for progressive loading, movement retraining or correction of the factors that contributed to the original injury. Its role is primarily to facilitate movement, reduce symptoms where appropriate, and allow the athlete to participate more effectively in the active rehabilitation programme.

The site-by-site techniques below are therefore not intended to represent a comprehensive manual therapy repertoire. They are a selection of techniques that may prove useful in this specific patient population and should be adapted according to the patient's presentation, irritability, stage of healing and response to treatment.

5.2.2 Self-Management

Athletes may be taught self-management techniques between sessions, including gentle self-massage of the forearm flexor musculature, mobility exercises, and carefully dosed use of tools such as massage balls or foam rollers.

Massage guns may also be used as part of an athlete's recovery routine, although evidence for their specific effectiveness in climbing-related tendon and pulley pathology remains limited — best considered an optional symptom-management strategy rather than a treatment for the underlying pathology.

5.A Finger & Hand Rehabilitation

Pulley Injury

Rehabilitation follows the grading established in Module 4A, with protective taping or splinting used throughout the loading progression — not as a substitute for graded reloading, but to keep the tendon approximated to the bone while capacity is rebuilt. Evidence on the biomechanical effectiveness of different taping methods has been mixed historically, with some studies questioning whether circumferential taping meaningfully reduces pulley strain and others supporting a protective effect; in practice, H-Tape is generally continued through the return-to-climbing phase

regardless, alongside — not instead of — a structured loading programme. A key element of successful taping is comfort: taping over a ganglion, for instance, can aggravate an injury rather than support the healing tissue.

Grade	Protection Phase	Return-to-Climbing Timeline	Protective Taping Duration
I–II	H-Tape or splint during loaded activity; early graded motion.	Approximately 6–8 weeks.	Around 3 months on return to climbing.
III	Thermoplastic ring splint; immobilisation guidance individualised to presentation.	Approximately 3 months.	Around 6 months on return to climbing. Worn 23 hrs/day for the first 6 weeks.
IV	Surgical reconstruction generally indicated before rehab progression.	Individualised, post-surgical timeline (commonly several months).	As directed post-operatively.

CLINICAL NOTE

Reloading after pulley injury should progress grip type deliberately: open-hand and slope-grip positions (lower pulley load) before half-crimp, and half-crimp well before any return to full crimp, since full crimp reintroduces the highest mechanical stress on the healing A2 region (Module 2, Section 2.5).

Flexor Tenosynovitis

Given its favourable natural history (Module 3), management centres on relative load modification — reducing training volume/intensity rather than complete cessation — alongside the same staged loading principles above. Climbers can generally continue modified training throughout recovery, with a graded return to prior volume as symptoms settle. It is, however, important to impress on the athlete that getting symptoms under control early is essential, as tenosynovitis can linger for up to a year, requiring several cortisone injections to settle.

Finger Joint Capsulitis / Synovitis

The published rehabilitation framework for this presentation follows four stages: unloading the irritated joint, restoring mobility, rebuilding strength, and correcting the movement or gripping fault that contributed to the injury. In the published case example, this progressive approach produced substantial pain and function improvement at six weeks, with continued improvement out to twelve months and a full return to prior climbing grade.

In lingering cases despite optimal load management, referral for an intra-articular cortisone injection could be required.

5.A.1 Treatment Modalities: Finger & Hand

Manual therapy in finger injuries should be used carefully, particularly during the early stages of pulley injury, flexor tendon pathology and acute joint irritation. The objective is not to mechanically "break up" healing tissue, but to address relevant restrictions or symptoms while allowing progressive tissue loading to continue.

Soft tissue techniques

Soft tissue work may be directed towards the flexor tendons and forearm flexor musculature where increased tone, tenderness or restricted tissue excursion is identified — particularly relevant in climbers with high volumes of gripping, crimping and finger-flexor loading.

For climbers with chronic pulley or tendon-sheath symptoms, treatment may also be directed around areas of localised thickening or restriction. Where nodular or scar-like tissue is present, grade treatment according to the stage of healing and the athlete's loading programme, and avoid aggressive work directly over a healing pulley or tendon.

Joint mobilisation

Accessory mobilisation of the PIP and other finger joints may be useful where assessment identifies a loss of joint mobility contributing to impaired movement or loading, graded according to irritability and aimed at restoring accessory range to support normal active movement — not simply increasing passive range.

5.B Wrist Rehabilitation

- **TFCC injury:** activity modification to avoid loaded ulnar deviation (sidepulls, underclings), wrist bracing or taping during early return to climbing, and grip/forearm strengthening; refer for orthopaedic opinion if conservative management fails to progress, given the limited standalone accuracy of the clinical tests discussed in Module 4B. A brace called the Wrist Widget has been suggested as a means of reducing load on the TFCC during rehabilitation, protecting it and thereby decreasing pain.
- **Hamate hook stress/fracture:** a period of stress reduction and activity modification, with conservative management reported to allow most climbers to return to their prior climbing level within roughly two to three months; hook resection is reserved for cases that fail conservative management.
- **De Quervain's tenosynovitis:** thumb spica bracing and activity modification acutely, followed by graded loading of the APL/EPB tendons, alongside technique correction — particularly reducing reliance on heavy vertical pinch grips and closed crimping with the thumb held in abduction/extension.

5.B.1 Treatment Modalities: Wrist

Manual therapy for wrist injuries should be guided by the specific structure involved and the direction of movement or loading that reproduces symptoms.

Soft tissue techniques

For ulnar-sided wrist presentations, soft tissue techniques may be directed towards the flexor carpi ulnaris (FCU) and surrounding forearm musculature where increased tone or tenderness is identified, alongside modification of provocative positions — particularly loaded ulnar deviation, sidepulls and underclings — where these reproduce symptoms.

Joint mobilisation

Where restriction in forearm or wrist movement is identified, appropriately graded accessory mobilisation may be used to facilitate movement. Forearm supination mobilisation may be particularly relevant where restricted supination is contributing to altered wrist or upper-limb mechanics.

5.C Elbow Rehabilitation

Both medial and lateral elbow tendinopathy in climbers respond to the same staged loading approach described in Section 5.2, typically progressing from isometric holds through eccentric or combined concentric-eccentric loading to heavier, higher-load resistance work as tolerated. Eccentric loading protocols specifically have evidence of benefit in medial epicondylalgia, though current evidence does not clearly establish eccentric loading as superior to other structured loading approaches — the more important factor is progressive, tolerated loading rather than any single exercise type.

Climbing-specific technique correction is a necessary companion to loading: reducing over-gripping, avoiding a flexed-wrist grip position, and addressing postural faults such as excessive hip sag away from the wall, all of which increase demand through the forearm flexor-pronator or extensor origin.

5.C.1 Treatment Modalities: Elbow & Forearm

Manual therapy can be particularly useful as an adjunct to rehabilitation in climbers presenting with elbow and forearm pain, where high repetitive gripping and forearm loading may be accompanied by increased muscle tone, restricted joint movement or altered neural mobility.

Soft tissue techniques

Soft tissue treatment may be directed towards the forearm flexor-pronator group, including pronator teres, where increased tone, tenderness or restricted movement is identified. For lateral elbow presentations, treatment may also be directed towards the extensor musculature where clinically indicated.

Elbow joint mobilisation

An anterior-to-posterior (AP) mobilisation of the radial head may be considered where assessment identifies a clinically relevant restriction in elbow or forearm movement, used alongside active movement and loading rather than as a stand-alone intervention.

Neural mobility

Neural mechanosensitivity should be considered when symptoms suggest involvement of the median, radial or ulnar nerves. Neural mobilisation may be used where appropriate, performed as controlled movements rather than aggressive stretches, aiming to improve neural mobility and tolerance to movement without provoking a prolonged increase in symptoms.

Neural interface treatment

Where the clinical examination suggests restriction at a neural interface, soft tissue treatment may be directed towards relevant surrounding structures — pectoralis minor, biceps and supinator for median nerve-related presentations, or triceps, FCU and brachioradialis for ulnar or radial nerve-related symptoms, depending on the anatomical site and presentation. Treatment should always be based on the suspected site of neural irritation rather than a generic "nerve release" approach.

Self-management

Athletes may be taught forearm self-massage using their hands, massage balls or other suitable tools, with foam rollers and massage guns available as optional strategies. Counterforce braces may help some athletes during the return-to-loading phase, used alongside — not instead of — progressive loading to restore tissue capacity.

5.D Shoulder Rehabilitation

Scapular control is the foundation of shoulder rehabilitation in climbers: strengthening the serratus anterior and lower/middle trapezius to restore coordinated scapular upward rotation during overhead reaching reduces subacromial crowding and is standard practice across the impingement and rotator cuff tendinopathy presentations covered in Module 4D. Climbing-specific reloading should reintroduce the "active hang" (engaging the scapula rather than hanging passively from the joint capsule), rebalance the sport's inherently pull-dominant loading with regular antagonist/push-pattern and external rotation work, and reintroduce dynamic and overhead-dominant movement only once basic strength and control are established.

5.D.1 Treatment Modalities: Shoulder & Thoracic Spine

The climbing shoulder is exposed to high volumes of pulling, overhead positioning, scapular control and sustained loading. Manual therapy may therefore be directed not only at the glenohumeral joint and shoulder musculature, but also at the thoracic spine where mobility restrictions are often identified. A stiff thoracic spine, particularly reduced thoracic extension and rotation, may contribute to altered scapulothoracic mechanics and should be assessed as part of the overall shoulder examination.

Soft tissue techniques

- Pectoralis minor
- Anterior deltoid
- Posterior shoulder tissues
- Relevant rotator cuff musculature
- Other surrounding tissues identified as contributing to movement restriction

Pectoralis minor treatment may be particularly useful where restriction is affecting scapular positioning or upward rotation, while posterior shoulder soft tissue techniques — including the posterior capsule region — may be considered where restricted internal rotation or posterior stiffness is present.

Glenohumeral joint mobilisation

AP glenohumeral mobilisations may be considered where assessment identifies a clinically relevant restriction in shoulder movement. Where internal rotation is limited, treatment should address the contribution of the posterior shoulder and thoracic spine alongside the glenohumeral joint itself.

Thoracic mobilisation

Thoracic mobility should be assessed in climbers presenting with shoulder dysfunction, particularly where extension or rotation appears restricted. Appropriately graded thoracic mobilisation can then be integrated with active thoracic mobility, scapular control and shoulder strengthening exercises.

Proprioception and open-chain control

Shoulder rehabilitation should not rely exclusively on strength — proprioceptive retraining and controlled open-chain exercises using free weights can be useful for restoring dynamic control, progressing from low-load movement to more demanding overhead and sport-specific positions as tolerance improves.

Self-management

Climbers may be taught self-management for the pectoral and posterior shoulder musculature, including controlled self-massage, mobility exercises and stretching where indicated, aimed at improving movement tolerance and supporting participation in the rehabilitation programme.

CLINICAL NOTE

Functional and sport-specific outcomes after surgical rotator cuff repair in climbers have been reported as good, supporting a realistic and encouraging conversation with patients who are facing surgical management — though, as with the general population, outcomes are better when post-operative rehabilitation is structured and criteria-based rather than purely time-based.

5.E Knee Rehabilitation

Conservative management with early functional physiotherapy produces good to excellent outcomes in the majority of technique-related knee injuries (heel hook, high step, drop knee), even where structural damage such as a partial meniscal tear is present — surgery is reserved for a minority of cases. Rehabilitation should combine standard structure-specific loading (quadriceps/hamstring strengthening, meniscal-friendly range progression, ITB-directed work where relevant) with technique-specific reintroduction: heel hook loading progressions from partial to full body weight, and drop knee/high step practice at controlled heights before returning to unrestricted movement on the wall.

Ground-fall knee injuries, being higher-energy and more frequently multi-structural, more often require surgical management and a correspondingly longer, structure-led rehabilitation pathway rather than a climbing-technique-specific one.

5.E.1 Treatment Modalities: Knee

Manual therapy should be considered where the clinical examination identifies a specific mobility restriction or soft tissue limitation contributing to impaired function — particularly relevant following periods of reduced activity or immobilisation, or where local soft tissue restriction is affecting climbing-specific positions such as high steps, drop knees or heel hooks.

Soft tissue techniques

Soft tissue treatment may be directed towards relevant musculature around the knee and hip where increased tone, tenderness or movement restriction is identified, remaining structure-specific and avoiding aggressive work over an acutely injured or inflamed structure.

Joint mobilisation

Appropriately graded knee joint mobilisation may be used where a clinically relevant restriction in range of motion is identified, combined with active range-of-motion exercises, progressive quadriceps and hamstring strengthening, proprioceptive training and climbing-specific movement retraining.

5.F Ankle & Foot Rehabilitation

- **Lateral ankle sprain:** functional rehabilitation (early protected mobilisation and proprioceptive training) is preferred over prolonged immobilisation, in line with general sports medicine practice; footwear and technique factors (shoe fit,

smearing/edging control) are worth addressing alongside standard sprain rehabilitation given their contribution to the original mechanism.

- **Chronic shoe-related overuse** (retrocalcaneal bursitis, Achilles tendinopathy, symptomatic hallux valgus): rehabilitation alongside footwear modification is central here — addressing the tissue without addressing shoe fit and sizing is unlikely to resolve a mechanically driven overuse presentation.
- **Calcaneus/talus fracture and osteochondral talar lesions:** managed via standard orthopaedic fracture guidelines, with surgical fixation common in climbers given the high activity demands of the population; return to climbing is guided by fracture healing and orthopaedic clearance rather than a climbing-specific timeline.

5.F.1 Treatment Modalities: Ankle & Foot

Manual therapy may be useful where restricted ankle or foot mobility contributes to altered movement mechanics during edging, smearing, high steps or other climbing-specific positions.

Joint mobilisation

Where appropriate, accessory mobilisation may address clinically relevant restrictions in ankle or foot movement, selected according to the specific restriction identified on assessment.

Soft tissue techniques

Soft tissue treatment may be directed towards the calf complex and other relevant tissues where increased tone or restriction contributes to impaired ankle mobility or movement.

Self-management

Athletes may be taught calf and foot self-massage, mobility exercises and progressive loading strategies, with footwear modification remaining an important component where shoe-related overuse is contributing to symptoms.

5.3 Clinical Integration of Manual Therapy

Manual therapy should be integrated into the overall rehabilitation process rather than delivered as an isolated treatment block. A useful clinical sequence is:

- Identify the primary tissue or structure involved.
- Determine whether pain, stiffness, reduced tissue mobility or neural mechanosensitivity is limiting movement or loading.
- Select the most appropriate manual or soft tissue technique based on the examination findings.
- Apply the technique at an appropriate grade and dosage for the patient's irritability.
- Reassess the relevant movement, symptom or functional limitation.
- Use any improvement to facilitate active exercise, movement retraining or progressive loading.
- Monitor the patient's response over the following 24 hours.

CLINICAL PRINCIPLE

Manual therapy is an adjunct to active rehabilitation — use it when it addresses an identifiable clinical impairment and the response can be integrated into active treatment, not simply because a technique is available or a body region is commonly treated with it.

For climbers, consider assessing:

- Local soft tissue restriction around the injured structure
- Forearm flexor and extensor tone
- Finger and PIP joint mobility
- Elbow and radial-head mobility
- Forearm pronation and supination
- Glenohumeral internal rotation
- Posterior shoulder and pectoralis minor restriction
- Thoracic extension and rotation
- Neural mobility and neural interfaces where clinically indicated
- Lower-limb mobility relevant to high steps, heel hooks and drop knees

Remember: follow a successful manual therapy intervention with an active movement or loading intervention that reinforces the change obtained during treatment.

5.3.1 Position- and Mechanism-Specific Rehabilitation

Throughout this module, loading progressions are described in terms of structure and stage — but equally important is ensuring that both manual therapy and active rehabilitation are matched to the specific position and mechanism under which the injury actually occurred, not simply the structure involved in general. This reflects a basic training principle — specific adaptation to imposed demands (SAID) — that applies as much to climbing rehabilitation as to any other sport: tissue adapts most effectively to the load, range, and position it is actually asked to tolerate, and a generic strengthening or loading programme performed in a neutral or unrelated position may leave a genuine gap between clinic-based rehabilitation and the climbing-specific demand the tissue will eventually face.

A useful example is biceps femoris rehabilitation following a heel hook injury. A hard or aggressive heel hook loads the hamstring in a position combining hip and knee flexion, hip external rotation, tibial external rotation and active hip extension and knee flexion effort against resistance, as the climber actively pulls the hips toward the wall through the heel. Generic hamstring strengthening — standard prone or seated hamstring curls performed with the hip in a neutral position, for example — does not reproduce this combined position and loading pattern, and completing this alone before returning to climbing risks leaving the tissue under-prepared for the exact demand that caused the original injury. Rehabilitation should instead progress toward loading the hamstring specifically in hip- and knee-flexed, hip-externally-rotated and tibia-externally-rotated positions that approximate the heel hook itself — for example, single-leg positions combining hip flexion and external rotation with resisted hip extension — before progressing to controlled heel hook loading on a wall or box at increasing body-weight percentages, as already

outlined in Section 5.E.

The same logic extends to manual therapy: where soft tissue treatment is directed towards a muscle that is symptomatic in a specific climbing position, treating that tissue in or near a lengthened/shortened, position-specific state — rather than a generic neutral position — may more directly address the tissue's actual functional demand, provided this is tolerated by the patient's irritability.

This principle recurs elsewhere in this module even where not explicitly restated: grip-type-specific reloading for pulley injuries (Section 5.A), progressive reintroduction of the pronated, gripped elbow position for medial epicondylalgia (Section 5.C), and graded reintroduction of the overhead, dynamic shoulder position (Section 5.D) are all applications of the same underlying idea — rehabilitation should ultimately be specific not just to the injured structure, but to the position and movement pattern that will be demanded of it on the wall.

5.4 Climbing-Specific Exercise Selection & Reintroduction to the Wall

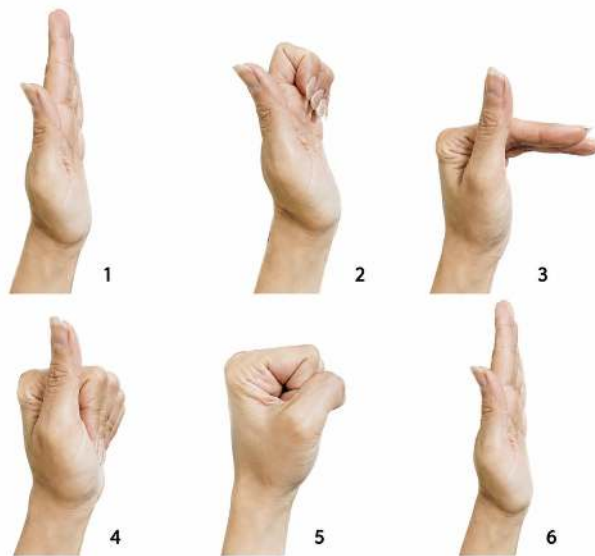
Across every site in this module, a consistent principle applies when reintroducing climbing itself: volume before intensity, controlled movement before dynamic movement, and general climbing before specific training tools. In practice this means restoring general movement on easy terrain before reintroducing technique-specific loading (heel hooks, crimping, dynamic moves), and delaying structured finger training tools such as hangboards and campus boards until later in the rehabilitation process — consistent with the slower adaptation rate of tendon and pulley tissue relative to muscle established in Module 2.

This approach allows for a much sooner return to climbing than one would think in most cases. Completely "benching" someone for more than 10 days is the exception rather than the rule.

Climbing Rehabilitation Exercise Library

A working set of exercise ideas to draw on when building a rehabilitation programme for each region covered in this module — starting points to select from and progress, not a fixed prescription.

Finger & Hand



Tendon Glides

A sequence of finger positions — straight, hook, fist, and straight fist — moved through in order to encourage full excursion of the flexor tendons through their sheath. Typically used early in rehabilitation to maintain tendon mobility and reduce adhesion risk while heavier loading is still restricted.

Loaded Tendon Glides

A gliding sequence performed against light resistance from a band attached to a hangboard, adding a small strengthening component once basic tendon glides are pain-free and well tolerated. (Video link supplied for this exercise could not be verified as a real, findable video and has been omitted — please confirm the correct link.)



Hangs (general term)

Hanging from a hangboard edge in a fixed grip position (usually open-hand or half-crimp, rarely full crimp given the joint stress). “Hangs” on its own is shorthand; the specific protocol below is what actually determines the training effect.



No Hangs

A finger strength training method where the climber's feet stay on the floor and they pull upward against a portable hold or block attached to a weight stack, rather than suspending their full bodyweight. This allows finer load control than a bodyweight hang — useful both for lighter climbers who can't easily reduce load below full bodyweight, and for early-stage rehab where bodyweight itself is too much.



Max Hangs

Near-maximal load for a short duration — typically 7–10 seconds, at or close to 60–90% of the heaviest weight the climber can hold on a given edge. Load can be bodyweight alone, bodyweight with a percentage taken off via a pulley and counterweight system, or bodyweight plus added weight. Full recovery between attempts, often 3+ minutes.

Repeaters

Short hangs repeated with brief rest within the same set — the classic format is “7 seconds on, 3 seconds off,” repeated for 6–7 reps. This builds local muscular endurance and the capacity to sustain repeated moderate loads across a boulder problem or route, rather than raw peak force.

Density Hangs

Longer continuous holds — typically 20–60+ seconds — at meaningfully lower relative intensity than max hangs or repeaters, usually on a larger, more comfortable edge. The goal is building local tissue capacity rather than peak force.

Recruitment Pulls

A brief, explosive pull (roughly 1–3 seconds), grabbing the edge and pulling in hard for a very short burst. The goal is neural: recruiting high-threshold motor units without the sustained time-under-tension that a max hang or repeater set involves.

Hangboard Variables

The following variables are adapted depending on the stage of rehab, stage of training, and stage of athlete development:

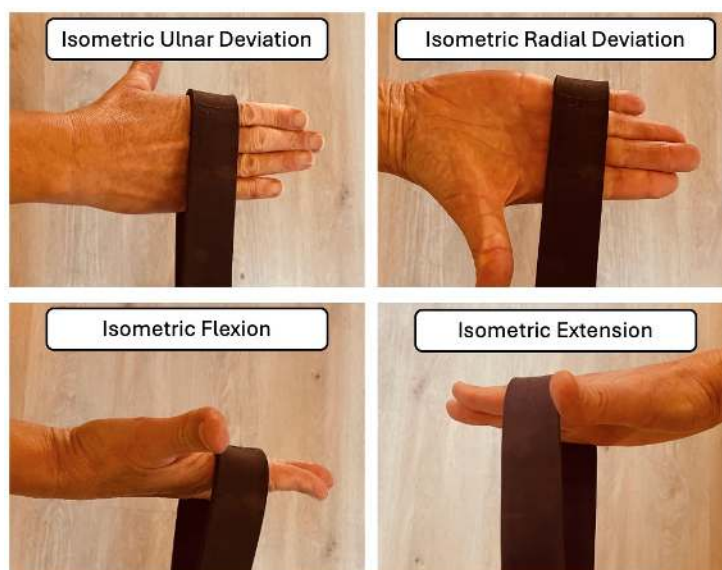
Variable	Description & Guidance
Edge depth	A 20mm edge is a standard reference point. Shallower edges increase difficulty; deeper edges reduce it.
Grip position	Open hand preferred for training. Half crimp acceptable. Full crimp should be avoided on hangboards.
Duration of hang	5–10 seconds for max strength; 20–45 seconds for endurance-oriented protocols.
Rest between sets	Max strength: 2–3 min. Endurance: 30–60 sec (intentional incomplete recovery).
Load	Bodyweight only for beginners; add or reduce weight to adjust difficulty as technique allows.
Number of sets	4–6 sets for strength focus; 2–3 sets for maintenance.

CLINICAL NOTE — BOARD CLIMBING AND FINGER TRAINING

Board climbing places intense, dynamic load through the same finger flexor tendons and pulleys targeted by dedicated finger training. Combining board climbing with dedicated finger training in the same session compounds load without adequate recovery — separate them across the week rather than stacking them in one session, and judge total weekly finger load rather than any single session's load.

Off the Wall Rehabilitation Ideas

Wrist Rehabilitation Ideas



Wrist Isometrics

Sustained, static contraction of the wrist flexors or extensors against resistance in a fixed position — a low-irritability starting point for wrist loading, matching the isometric stage of the staged loading model in Section 5.2.

Wrist Resistance Exercise with Weight

Dynamic strengthening of the wrist flexors, extensors, and radial/ulnar deviators through range — using resistance bands, a wrist roller, or free weights such as dumbbell wrist curls. Progressed once isometric loading is well tolerated.

Elbow Rehabilitation Ideas

Wrist Isometrics, and Wrist Resistance Exercise with Bands, Sticks, or Weight

The same exercises described under Wrist Rehabilitation Ideas above apply directly here too — the forearm flexor-pronator and extensor muscles trained through wrist loading originate at the medial and lateral epicondyle respectively.

Median/Radial/Ulnar Nerve Slides

Controlled neural mobilisation exercises that move a targeted nerve through its available range relative to surrounding tissue, used where clinical examination suggests altered neural mechanosensitivity is contributing to elbow or forearm symptoms.

Watch demonstration: <https://youtu.be/ktOw-6irlYs>

Shoulder Rehabilitation Ideas

Scapular control and pull/push balance can be built with a small set of accessible exercises:

Resistance Band Pull Apart

Hold a band at chest height with arms extended, then pull it apart by squeezing the shoulder blades together. Targets the middle trapezius, rhomboids, rear deltoids and rotator cuff.

Ring Rows

A bodyweight pulling exercise: from a straight-body lean, pull up by driving the elbows down and back until the chest reaches the rings. Loads latissimus dorsi, rhomboids, middle trapezius and biceps.

Resistance Band Face Pull

Pull a band towards the face by leading with the elbows, keeping them high, squeezing the shoulder blades together at end range. Targets the rear deltoids, rotator cuff and lower/middle trapezius.

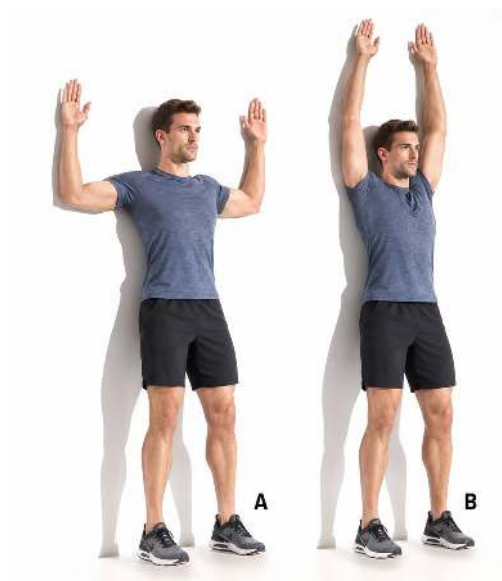
Turkish Get-Up

A full-body movement that takes the athlete from lying on the floor to standing while keeping a weight locked out overhead. Builds shoulder stability, core control and proprioception under a dynamically changing load.



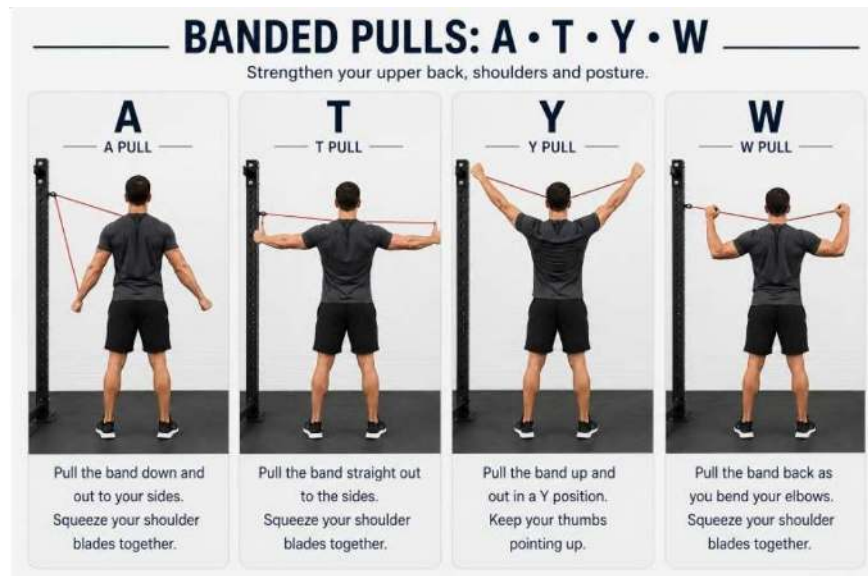
Wall Push Up, Serratus Punch, Push Up Plus

A progression targeting serratus anterior specifically — each involves protracting the scapula at end-range against resistance, directly building the serratus anterior strength foundational to shoulder rehabilitation.



Scapular Wall Angel

Arms held against a wall in a “goal-post” position, sliding overhead while maintaining contact with the wall throughout. Trains scapular upward rotation and posterior tilt with direct feedback on positioning.



ATYW Banded Pulls

A series of banded arm positions — forming an A, T, Y, and W shape — that progressively target different portions of the trapezius, rhomboids, and rotator cuff.



Active Hang

Keeping the scapulae actively engaged (depressed and controlled) rather than passively hanging from the joint capsule — progressed toward fuller passive hanging only once scapular control is established.

Turkish Get Up Versions and Overhead Carries

Full-body exercises that load the shoulder in a stable overhead position while the rest of the body moves beneath it (get-ups) or through space (carries). Builds sustained overhead shoulder stability.

Thoracic Extension and Rotation Versions





Mobility work targeting thoracic extension and rotation specifically, addressing the stiff thoracic spine contributing to altered scapulothoracic mechanics.

Climbing-Specific Overhead Reaching

The final-stage reintroduction of dynamic, overhead-dominant reaching patterns once basic strength and scapular control are established — the last step before returning to unrestricted climbing movement.

Knee Rehabilitation Ideas

Strength

Image coming soon

Squat Progression (2-leg → 1-leg, range progressions)

General lower-limb strength base; single-leg progression carries over directly to climbing's near-constant unilateral weight-bearing.

Image coming soon

Hamstring Progression (bridge: 2-leg → 1-leg → ball → rings → band)

Direct heel hook relevance — builds the hip-extension/knee-flexion strength the hamstring needs to actively pull the hips toward the wall.

Image coming soon

Bulgarian Split Squat

Single-leg strength with the rear leg elevated, loading the front leg through a longer range — mirrors the demand of a high step or mantel.

Image coming soon

Cossack Squat

Lateral squat pattern building hip strength and mobility in a wide, loaded stance — relevant to smearing, flagging, and wide stemming.

Image coming soon

Step Up Progression (forward → sideways → height → plyometrics)

Direct high step relevance — progressively loads the exact movement pattern a high step demands.

Image coming soon

Proprioception

Builds joint position sense and balance control — relevant given climbing's demand for precise foot placement.

Range of Motion

Image coming soon

Drop Knee

Hip internal rotation and adductor range specifically — the mobility a drop knee position requires to rotate the hip toward the wall.

Image coming soon

Froggie Stretch

Hip external rotation and groin/adductor range — supports the wide hip-opening positions demanded by drop knees, high steps, and heel hooks.

Ankle & Foot Rehabilitation Ideas

Image coming soon

Knee-to-Wall Mobility

Targets ankle dorsiflexion range, allowing the climber to drop their heel and maximise rubber contact with the wall in slab positions.

Image coming soon

Calf Raises

Build the strength and endurance needed to sustain weight on the toes or generate force through a toe-down position.

Image coming soon

Toe Hook — Resisted Dorsiflexion with a Band

Directly trains the toe hook mechanism using a band to resist ankle dorsiflexion.

Image coming soon

Big Toe Flexion with a Band

Builds toe-down precision and force transfer through small edges.

Image coming soon

Proprioception and Edging Footwork off the Wall

Practised using blocks or volumes fitted with holds, allowing controlled practice of foot placement and weight transfer.

Shoe Sizing

Climbing shoes are typically sized significantly smaller than the climber's everyday shoe size — often three to four sizes down — to maximise sensitivity and precision of pressure transfer. This tighter fit increases the risk of ankle sprains, bunions and other toe deformities, and is worth factoring into any assessment of chronic foot pain in a climbing patient.

Summary

Rehabilitation of climbing injuries follows familiar physiotherapy loading principles — protection, graded reloading, and a staged return to sport-specific demand — applied through a climbing-specific lens: grip type progression for the fingers, technique correction for the elbow and shoulder, and movement-specific reintroduction for the knee and ankle. With treatment established, Module 6 addresses the final stage of the pathway: grading readiness for return to sport, managing load progression by discipline, and monitoring for re-injury risk.

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